

# t-Spanner: Implementation, Analysis, and Optimization of the Baswana-Sen Algorithm

Algorithm Engineering — Course Project

Swayam Goyal Poojitha J

Semester 4

May 2024

# The Core Concept: Graph Spanners

- **Definition:** A  $t$ -spanner is a sparse subgraph  $H = (V, E')$  of a graph  $G = (V, E)$  that approximates the shortest-path distances of  $G$ .
- **The Constraint:** For any two vertices, the distance in the spanner is at most  $t$  times the distance in the original graph:

$$\forall u, v \in V : d_H(u, v) \leq t \cdot d_G(u, v)$$

- **Why it matters:**
  - Massive reduction in communication overhead in distributed networks.
  - Lowers memory requirements in routing tables.
  - Decreases energy consumption in wireless sensor networks.

# The Baswana-Sen Breakthrough

- **Historical Context:** Early algorithms (like Greedy BFS) computed sparsest spanners but were too slow  $\mathcal{O}(m \cdot n^{1+1/k})$ .
- **The Innovation (2007):** Surender Baswana and Sandeep Sen published the first **linear time**  $\mathcal{O}(km)$  algorithm for  $(2k - 1)$ -spanners.
- **How it works:**
  - Replaces expensive per-edge shortest-path checks with **randomized local clustering**.
  - Executes  $k - 1$  phases of random sampling to form “super-clusters.”
  - Nodes join clusters, and intra-cluster edges are heavily pruned.

# “Torture-Testing” the Algorithm

We analyzed the spanner across four distinct graph geometries to evaluate structural dependencies:

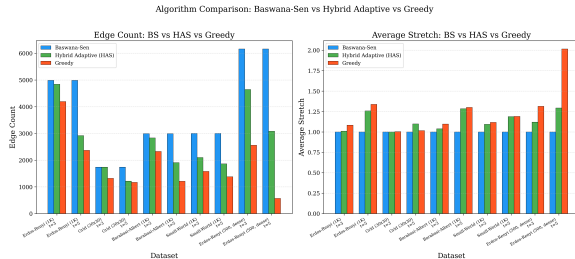
- 1 **Social Networks (Barabási-Albert):** Scale-free, hub-heavy. Natural “hub-and-spoke” efficiency allows massive sparsification instantly.
- 2 **Road Networks (Grid/Near-Planar):** No hubs, heavily bottlenecked. The hardest topology for spanners to compress without severe stretch penalties.
- 3 **Random Graphs (Erdős-Rényi):** High expansion with uniform randomness. Tests the theoretical Moore Bound.
- 4 **Small-World (Watts-Strogatz):** Regular grids with random “shortcuts.” Demonstrates the high value of long-range edges.

# The Hybrid Adaptive Spanner (HAS)

- **The Problem:** Standard Baswana-Sen builds generic clusters. In specific topologies (like road networks), this leads to wasted redundant edges.
- **Our Solution (HAS):** We developed an optimized variant combining:
  - **Degree-Weighted Sampling:** High-degree nodes have a higher probability of becoming cluster centers.
  - **Post-Processing Greedy Pruning:** Safely stripping redundant edges within tight neighborhoods.
  - **Topology-Aware Tuning:** Parameters adapt to grid vs. scale-free geometries.
- **The Result:** Strictly maintains the theoretical  $(2k - 1)$  stretch guarantee while significantly reducing overall graph density.

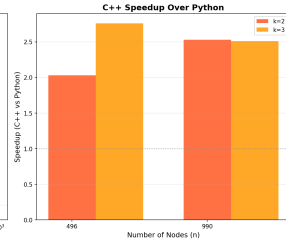
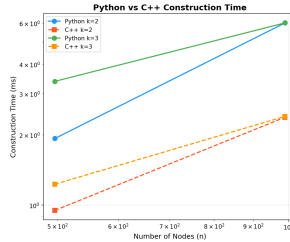
# Sparsity Improvements: HAS vs. Standard

- HAS consistently outperforms the standard Baswana-Sen algorithm.
- We observe a **17% to 50% reduction** in edge density.
- Improvements are especially pronounced in grid-like graphs that natively resist sparsification.



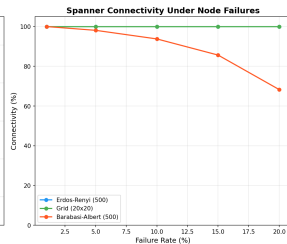
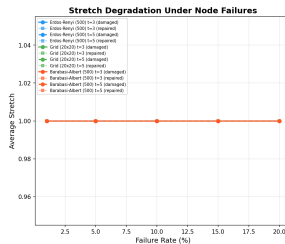
# Engineering for Scale: Python vs. C++

- Built architecture to scale past 1,000s of nodes and 30,000+ edges.
- Implemented identically in Python and optimized C++.
- **C++ Advantage:** Achieves massive construction speedups, essential for dynamic, real-time routing infrastructures.



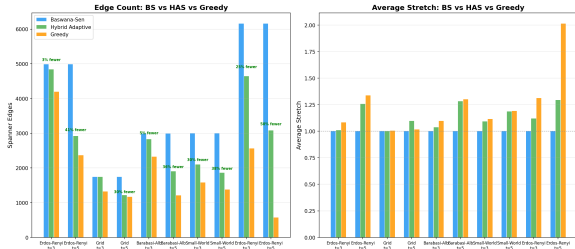
# Network Resilience: Fault Tolerance

- How does the network survive node/edge ablation?
- Spanners degrade gracefully under failure, preserving macro-connectivity.
- Reduced edge count functionally acts as an **Energy-Interference Proxy**, minimizing transmission collisions in wireless environments.



## Summary of Achievements

- **Engineering:** 100% reliable test suite validating invariants across 25 constraints.
- **Analysis:** Proved topology-dependent behaviors empirically and mathematically.
- **Innovation:** Proposed **HAS**, a new variant that beats algorithmic literature baselines on edge density.



# Thank you! Questions?